Pratim Mangaldas Dasude

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Bengaluru, Karnataka

Experience

AiLusion — Merosa Technologies Pvt. Ltd
Jan 2025 - Current
Machine Learning Engineer
Bengaluru, Karnataka

- Developed and fine-tuned **large-scale VLMs and diffusion models** using PyTorch, Flux-Dev, and CLIP-style embeddings, improving multimodal alignment, visual reasoning, and photorealistic garment–body draping.
- Built **scalable training and evaluation pipelines** with mixed-precision, distributed workflows, and automated ablations—reducing training latency by 30% and improving downstream retrieval/draping quality across 100+ SKUs.
- Applied **PEFT methods** such as LoRA, QLoRA, and DreamBooth for domain-specific adaptation of diffusion and VLM models, achieving higher visual fidelity and robust multimodal representations.

Dashboard — My Work

Highradius Technologies
July 2023 - November 2023
Data Science Intern
Bhubaneswar, Odisha

- Utilizing **XGBoost** and **LightGBM**, a Deduction classification model was constructed and optimized. Through **feature engineering** and **hyperparameter tuning**, the accuracy of the validity assessment was increased by **20%**, and processing time was decreased by **15%**.
- **Automated monitoring** and drift detection pipelines, reducing mispredictions by **30%** and streamlining operations for 100+ clients through iterative **agile** workflows.

Projects

EchoSeek | *Agentic RAG-Based Product Discovery Platform*

- Built a **multi-modal RAG pipeline** integrating **Llama-3.1** for reasoning and **NV-Embed-QA** for embeddings, leveraging **LangChain** to enable intelligent, real-time product retrieval.
- Orchestrated inference through **NVIDIA NIM** endpoints using **AWS Sagemaker** for conversational, context-aware recommendations. achieving **30%** faster retrieval and improved product relevance.

Skills: Python, LangChain, FastAPI, AWS SageMaker, Llama-3, RAG, NVIDIA NIM, Docker, Next.js

White Balance Regression Model | *EfficientNetV2-S*

- Built a **computer vision regression model** to estimate **Color Temperature (Kelvin)** and **Tint** from images, improving automatic white-balance correction accuracy by 25–30%.
- Fine-tuned **EfficientNetV2-S** using **multi-stage training** ($256\times256 \rightarrow 384\times384$) with **MSE loss** and **ColorJitter**, reducing prediction error by $\sim$20%.

Skills: PyTorch, Computer Vision, EfficientNet, Model Fine-Tuning

Order Amount Prediction | *Predicting amount for a B2B business*

- Conducted comprehensive **time series analysis** using **Random Forest** and **XGBoost**, achieving 75% prediction accuracy and enhancing forecasting precision by 15% through **ensemble learning**.
- Optimized model performance with advanced **feature engineering** and **hyperparameter tuning**, reducing Mean Absolute Error (MAE) by 20% on test data and strengthening prediction reliability.

Skills: Python, Matplotlib, Scikit-learn, Data Visualisation, Git.

Education

Kalinga Institute of Industrial Technologies , Bhubaneswar
B.Tech in Computer Science and Engineering (2020 - 2024) — CGPA - **9.15**

Julien Day School , Kalyani
ISC Board (2017 - 2019) — Percentage - **84.00**
ICSE Board (2015 - 2017) — Percentage - **81.83**

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C/C++, HTML/CSS, SQL, MongoDB, Bash, Matplotlib
Technologies: TensorFlow, PyTorch, Keras, scikit-learn, FastAPI, React.js, Next.js, Docker, FAISS, LangChain
Concepts: AI, ML, Neural Networks, Deep Learning, Computer Vision, Generative AI, LLMs, RAG, Agentic AI, MCP, LoRA Fine-Tuning, Hyperparameter Tuning, Feature Engineering, Model Training & Evaluation.
Tools:ComfyUI, Fooocus, Hugging Face (Transformers, Diffusers), Git, Jira, Google Colab, Firebase, MS Excel, Kaggle Datasets, AWS SageMaker, NVIDIA NIM, Postman

Research Paper

Speech Emotion (LSTM/CNN) Stress Detection (BERT-MLP) Multilingual Spam (BERT/LSTM)